

Commerce XI Maths App

Miscellaneous Exercise 1 - Detailed Answers with Explanations

Total questions: 46 | +2 correct | -1 wrong

This answer guide follows the question order used in the app.

Question 1 - Q1(i)

Write the following set in set builder form.

$\{10, 20, 30, 40, 50\}$

Options

- A. $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$
- B. $\{x / x = 10n, n \in \mathbb{W}, n < 5\}$
- C. $\{x / x = 5n, n \in \mathbb{N}, n \leq 10\}$
- D. $\{x / x = 10n, n \in \mathbb{N}, n < 5\}$

Correct answer: $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 2 - Q1(ii)

Write the following set in set builder form.

$\{a, e, i, o, u\}$

Options

- A. $\{x / x \text{ is vowel of English alphabets}\}$
- B. $\{x / x \text{ is consonant of English alphabets}\}$
- C. $\{x / x \text{ is a letter of the word "vowel"}\}$
- D. $\{x / x \text{ is an English alphabet}\}$

Correct answer: $\{x / x \text{ is vowel of English alphabets}\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \text{ is vowel of English alphabets}\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \text{ is vowel of English alphabets}\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 3 - Q1(iii)

Write the following set in set builder form.

$\{\text{Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}\}$

Options

- A. $\{x / x \text{ represents day of a week}\}$
- B. $\{x / x \text{ represents working day of a week}\}$
- C. $\{x / x \text{ represents month of a year}\}$
- D. $\{x / x \text{ represents weekend day}\}$

Correct answer: $\{x / x \text{ represents day of a week}\}$

Explanation and steps

1. The governing definition or property gives: $\{x / x \text{ represents day of a week}\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / x \text{ represents day of a week}\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 4 - Q2(i)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $A \cup B$.

Options

- A. $\{1, 2, 4, 6, 7, 10, 11\}$
- B. $\{4, 7\}$
- C. $\{1, 10\}$
- D. $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Correct answer: $\{1, 2, 4, 6, 7, 10, 11\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 4, 6, 7, 10, 11\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 4, 6, 7, 10, 11\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 5 - Q2(ii)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $B \cap C$.

Options

- A. $\{ \}$
- B. $\{3, 5, 8, 9, 12\}$

C. $\{2, 4, 6, 7, 11\}$

D. $\{4, 7\}$

Correct answer: $\{ \}$

Explanation and steps

1. The governing definition or property gives: $\{ \}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{ \}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 6 - Q2(iii)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $A - B$.

Options

A. $\{1, 10\}$

B. $\{2, 6, 11\}$

C. $\{4, 7\}$

D. $\{1, 2, 6, 10, 11\}$

Correct answer: $\{1, 10\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 10\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 10\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 7 - Q2(iv)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $B - C$.

Options

A. $\{2, 4, 6, 7, 11\}$

B. $\{ \}$

C. $\{3, 5, 8, 9, 12\}$

D. $\{2, 4, 6, 7, 11, 12\}$

Correct answer: {2, 4, 6, 7, 11}

Explanation and steps

1. The governing definition or property gives: {2, 4, 6, 7, 11}.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is {2, 4, 6, 7, 11}.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 8 - Q2(v)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $A \cup B \cup C$.

Options

- A. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
- B. {1, 2, 4, 6, 7, 10, 11}
- C. {3, 4, 5, 7, 8, 9, 12}
- D. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}

Correct answer: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

Explanation and steps

1. The governing definition or property gives: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 9 - Q2(vi)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.
Write $A \cap (B \cup C)$.

Options

- A. {4, 7}
- B. { }
- C. {1, 4, 7, 10}
- D. {1, 10}

Correct answer: {4, 7}

Explanation and steps

1. The governing definition or property gives: $\{4, 7\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{4, 7\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 10 - Q3

In a survey of 425 students in a school, it was found that 115 drink apple juice, 160 drink orange juice and 80 drink both apple as well as orange juice.
How many drink neither apple juice nor orange juice?

Correct answer: 230

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 230.
3. Therefore, the required answer is 230.

Question 11 - Q4

In a school there are 20 teachers who teach Mathematics or Physics, of these, 12 teach Mathematics and 4 teach both Physics and Mathematics.
How many teachers teach Physics?

Correct answer: 12

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 12.
3. Therefore, the required answer is 12.

Question 12 - Q5(i) $A \times A$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.
State the elements of $A \times A$.

Options

- A. $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
- B. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$
- C. $\{(1, 1), (2, 2), (3, 3)\}$
- D. $\{(1, 2), (2, 1), (2, 3), (3, 2)\}$

Correct answer: $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

Explanation and steps

1. The governing definition or property gives: $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 13 - Q5(i) $A \times B$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $A \times B$.

Options

- A. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$
- B. $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$
- C. $\{(1, 2), (2, 2), (3, 2)\}$
- D. $\{(1, 4), (2, 4), (3, 4)\}$

Correct answer: $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

Explanation and steps

1. The governing definition or property gives: $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 14 - Q5(i) $B \times A$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $B \times A$.

Options

- A. $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$
- B. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$
- C. $\{(2, 1), (2, 2), (2, 3)\}$
- D. $\{(4, 1), (4, 2), (4, 3)\}$

Correct answer: $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$

Explanation and steps

1. The governing definition or property gives: $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 15 - Q5(i) $B \times B$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $B \times B$.

Options

- A. $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$
- B. $\{(2, 4), (4, 2)\}$
- C. $\{(2, 2), (4, 4)\}$
- D. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

Correct answer: $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$

Explanation and steps

1. The governing definition or property gives: $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 16 - Q5(i) Intersection

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

Find $(A \times B) \cap (B \times A)$.

Options

- A. $\{(2, 2)\}$
- B. $\{(1, 2), (2, 1)\}$
- C. $\{(2, 4), (4, 2)\}$
- D. $\{ \}$

Correct answer: $\{(2, 2)\}$

Explanation and steps

1. The governing definition or property gives: $\{(2, 2)\}$.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $\{(2, 2)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 17 - Q5(ii)

If $A = \{-1, 1\}$, find $A \times A \times A$.

Options

- A. $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$
- B. $\{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$
- C. $\{(-1, -1, -1), (-1, 1, 1), (1, -1, 1), (1, 1, -1)\}$
- D. $\{(-1, -1, -1), (1, 1, 1)\}$

Correct answer: $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$

Explanation and steps

1. The governing definition or property gives: $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 18 - Q6(i)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R = \{(1, 4), (1, 5), (1, 6)\}$.

Which statement is correct?

Options

- A. R is a relation from A to B
- B. R is not a relation because 2 and 3 have no images
- C. R is not a relation because 1 has three images
- D. R is a relation from B to A

Correct answer: R is a relation from A to B

Explanation and steps

1. The governing definition or property gives: R is a relation from A to B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is R is a relation from A to B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 19 - Q6(ii)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R = \{(1, 5), (2, 4), (3, 6)\}$.
Which statement is correct?

Options

- A. R is a relation from A to B
- B. R is not a relation because the pairs are not in $A \times B$
- C. R is a relation from B to A
- D. R is not a relation because each first component is different

Correct answer: R is a relation from A to B

Explanation and steps

1. The governing definition or property gives: R is a relation from A to B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is R is a relation from A to B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 20 - Q6(iii)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R = \{(1, 4), (1, 5), (3, 6), (2, 6), (3, 4)\}$.
Which statement is correct?

Options

- A. R is a relation from A to B
- B. R is not a relation because some first components repeat
- C. R is a relation from B to A
- D. R is not a relation because 2 and 3 have the same image

Correct answer: R is a relation from A to B

Explanation and steps

1. The governing definition or property gives: R is a relation from A to B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is R is a relation from A to B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 21 - Q6(iv)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R = \{(4, 2), (2, 6), (5, 1), (2, 4)\}$.
Which statement is correct?

Options

- A. R is not a relation from A to B
- B. R is a relation from A to B
- C. R is empty relation from A to B
- D. R is the universal relation from A to B

Correct answer: R is not a relation from A to B

Explanation and steps

1. The governing definition or property gives: R is not a relation from A to B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is R is not a relation from A to B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 22 - Q7 Domain

Determine the domain of the following relation.

$$R = \{(a, b) / a \in \mathbb{N}, a < 5, b = 4\}$$

Options

- A. $\{1, 2, 3, 4\}$
- B. $\{0, 1, 2, 3, 4\}$
- C. $\{4\}$
- D. $\{1, 2, 3, 4, 5\}$

Correct answer: $\{1, 2, 3, 4\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 3, 4\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 3, 4\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 23 - Q7 Range

Determine the range of the following relation.

$$R = \{(a, b) / a \in \mathbb{N}, a < 5, b = 4\}$$

Options

- A. {4}
- B. {1, 2, 3, 4}
- C. {0, 1, 2, 3, 4}
- D. {1, 2, 3, 4, 5}

Correct answer: {4}

Explanation and steps

1. The governing definition or property gives: {4}.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is {4}.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 24 - A1.1

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3\}$, and $B = \{3, 4\}$.
Find $(A \cup B)'$.

Options

- A. {5, 6, 7, 8, 9, 10}
- B. {1, 2, 3, 4}
- C. {3}
- D. {1, 2, 4, 5, 6, 7, 8, 9, 10}

Correct answer: {5, 6, 7, 8, 9, 10}

Explanation and steps

1. The governing definition or property gives: {5, 6, 7, 8, 9, 10}.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is {5, 6, 7, 8, 9, 10}.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 25 - A1.2

Let A and B be nonempty subsets of a universal set.
Which condition makes $A \cup B = A \cap B$?

Options

- A. $A = B$
- B. $A \cap B = \{ \}$

- C. $A \subset B$ only
- D. $B \subset A$ only

Correct answer: $A = B$

Explanation and steps

1. The governing definition or property gives: $A = B$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A = B$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 26 - A1.3

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Which statement is correct?

Options

- A. $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$
- B. $A \times B = B \times A$, but $n(A \times B) \neq n(B \times A)$
- C. $A \times B = A \cup B$
- D. $n(A \times B) = n(A) + n(B)$

Correct answer: $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$

Explanation and steps

1. The governing definition or property gives: $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 27 - A1.4

What conclusion will you draw about two sets A and B if $A \subseteq B$ and $B \subseteq A$?

Options

- A. $A = B$
- B. $A \cap B = \{ \}$
- C. A is a proper subset of B
- D. B is a proper subset of A

Correct answer: $A = B$

Explanation and steps

1. The governing definition or property gives: $A = B$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A = B$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 28 - A1.5

If $A = \{-1/2, 2/3\}$, which set builder form represents A?

Options

- A. $\{x / 6x^2 - x - 2 = 0\}$
- B. $\{x / 6x^2 + x - 2 = 0\}$
- C. $\{x / 6x^2 - x + 2 = 0\}$
- D. $\{x / 3x^2 - x - 2 = 0\}$

Correct answer: $\{x / 6x^2 - x - 2 = 0\}$

Explanation and steps

1. The governing definition or property gives: $\{x / 6x^2 - x - 2 = 0\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{x / 6x^2 - x - 2 = 0\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 29 - A1.6 Domain

For the relation $R = \{(1, 5), (2, 6), (3, 7), (4, 8)\}$, write the domain.

Options

- A. $\{1, 2, 3, 4\}$
- B. $\{5, 6, 7, 8\}$
- C. $\{1, 2, 3, 4, 5, 6, 7, 8\}$
- D. $\{1, 5\}$

Correct answer: $\{1, 2, 3, 4\}$

Explanation and steps

1. The governing definition or property gives: $\{1, 2, 3, 4\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1, 2, 3, 4\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 30 - A1.6 Range

For the relation $R = \{(1, 5), (2, 6), (3, 7), (4, 8)\}$, write the range.

Options

- A. $\{5, 6, 7, 8\}$
- B. $\{1, 2, 3, 4\}$
- C. $\{1, 2, 3, 4, 5, 6, 7, 8\}$
- D. $\{4, 8\}$

Correct answer: $\{5, 6, 7, 8\}$

Explanation and steps

1. The governing definition or property gives: $\{5, 6, 7, 8\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{5, 6, 7, 8\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 31 - A1.7(i)

Write the following set in Roster form.

A – set of all factors of 24.

Options

- A. $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$
- B. $A = \{1, 2, 3, 4, 6, 8, 12\}$
- C. $A = \{2, 3, 4, 6, 8, 12, 24\}$
- D. $A = \{1, 2, 3, 4, 6, 8, 12, 24, 48\}$

Correct answer: $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$

Explanation and steps

1. The governing definition or property gives: $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 32 - A1.7(ii)

Write the following set in Roster form.

B – set of all prime numbers less than 30.

Options

A. $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

B. $B = \{1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

C. $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$

D. $B = \{2, 3, 5, 7, 9, 11, 13, 17, 19, 23, 29\}$

Correct answer: $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

Explanation and steps

1. The governing definition or property gives: $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 33 - A1.7(iii)

Write the following set in Roster form.

C – set of all letters in the word “MATHEMATICS”.

Options

A. $C = \{M, A, T, H, E, I, C, S\}$

B. $C = \{M, A, T, H, E, I, C\}$

C. $C = \{M, A, T, H, E, I, C, S, M\}$

D. $C = \{M, A, T, H, E, I, S\}$

Correct answer: $C = \{M, A, T, H, E, I, C, S\}$

Explanation and steps

1. The governing definition or property gives: $C = \{M, A, T, H, E, I, C, S\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $C = \{M, A, T, H, E, I, C, S\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 34 - A1.8

In a survey of 400 students, it was found that 150 drink milk, 250 drink Tea, and 50 drink both.

How many drink neither Tea nor milk?

Correct answer: 50

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 50.
3. Therefore, the required answer is 50.

Question 35 - A1.9 Union

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A \cup B$.

Options

- A. $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$
- B. $\{1/6, 1/12\}$
- C. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$
- D. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

Correct answer: $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$

Explanation and steps

1. The governing definition or property gives: $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 36 - A1.9 Intersection

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A \cap B$.

Options

- A. $\{1/6, 1/12\}$
- B. $\{1/3, 1/6, 1/9, 1/12\}$
- C. $\{1/2, 1/4, 1/6, 1/8\}$
- D. $\{ \}$

Correct answer: $\{1/6, 1/12\}$

Explanation and steps

1. The governing definition or property gives: $\{1/6, 1/12\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1/6, 1/12\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 37 - A1.9 A – B

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A - B$.

Options

- A. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$
- B. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$
- C. $\{1/6, 1/12\}$
- D. $\{1/3, 1/6, 1/9, 1/12, 1/15, 1/18, 1/21\}$

Correct answer: $\{1/3, 1/9, 1/15, 1/18, 1/21\}$

Explanation and steps

1. The governing definition or property gives: $\{1/3, 1/9, 1/15, 1/18, 1/21\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1/3, 1/9, 1/15, 1/18, 1/21\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 38 - A1.9 B – A

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $B - A$.

Options

- A. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$
- B. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$
- C. $\{1/6, 1/12\}$
- D. $\{1/2, 1/4, 1/6, 1/8, 1/10, 1/12, 1/14\}$

Correct answer: $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

Explanation and steps

1. The governing definition or property gives: $\{1/2, 1/4, 1/8, 1/10, 1/14\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\{1/2, 1/4, 1/8, 1/10, 1/14\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 39 - A1.10 Complements

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find A' and B' .

Options

$$A. A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$$

$$B. A' = \{1, 2, 3\}, B' = \{6, 7, 8\}$$

$$C. A' = \{4, 5\}, B' = \{4, 5\}$$

$$D. A' = \{6, 7, 8\}, B' = \{1, 2, 3, 4, 5\}$$

Correct answer: $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$

Explanation and steps

1. The governing definition or property gives: $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 40 - A1.10 Intersection

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find $A \cap B$ and $n(A \cap B)$.

Options

$$A. A \cap B = \{4, 5\}, n(A \cap B) = 2$$

$$B. A \cap B = \{1, 2, 3\}, n(A \cap B) = 3$$

$$C. A \cap B = \{6, 7, 8\}, n(A \cap B) = 3$$

$$D. A \cap B = \{1, 2, 3, 4, 5, 6, 7, 8\}, n(A \cap B) = 8$$

Correct answer: $A \cap B = \{4, 5\}, n(A \cap B) = 2$

Explanation and steps

1. The governing definition or property gives: $A \cap B = \{4, 5\}, n(A \cap B) = 2$.

2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $A \cap B = \{4, 5\}$, $n(A \cap B) = 2$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 41 - A1.10 $A - B$

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Find $A - B$ and $n(A - B)$.

Options

A. $A - B = \{1, 2, 3\}$, $n(A - B) = 3$

B. $A - B = \{4, 5\}$, $n(A - B) = 2$

C. $A - B = \{6, 7, 8\}$, $n(A - B) = 3$

D. $A - B = \{1, 2, 3, 4, 5\}$, $n(A - B) = 5$

Correct answer: $A - B = \{1, 2, 3\}$, $n(A - B) = 3$

Explanation and steps

1. The governing definition or property gives: $A - B = \{1, 2, 3\}$, $n(A - B) = 3$.

2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $A - B = \{1, 2, 3\}$, $n(A - B) = 3$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 42 - A1.10 $B - A$

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Find $B - A$ and $n(B - A)$.

Options

A. $B - A = \{6, 7, 8\}$, $n(B - A) = 3$

B. $B - A = \{1, 2, 3\}$, $n(B - A) = 3$

C. $B - A = \{4, 5\}$, $n(B - A) = 2$

D. $B - A = \{4, 5, 6, 7, 8\}$, $n(B - A) = 5$

Correct answer: $B - A = \{6, 7, 8\}$, $n(B - A) = 3$

Explanation and steps

1. The governing definition or property gives: $B - A = \{6, 7, 8\}$, $n(B - A) = 3$.

2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $B - A = \{6, 7, 8\}$, $n(B - A) = 3$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 43 - A1.10 Union

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Find $A \cup B$ and $n(A \cup B)$.

Options

A. $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$

B. $A \cup B = \{4, 5\}$, $n(A \cup B) = 2$

C. $A \cup B = \{1, 2, 3, 6, 7, 8\}$, $n(A \cup B) = 6$

D. $A \cup B = \{1, 2, 3, 4, 5\}$, $n(A \cup B) = 5$

Correct answer: $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$

Explanation and steps

1. The governing definition or property gives: $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$.

2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 44 - A1.10(i)

If $n(A - B) = 3$, $n(A \cap B) = 2$, and $n(B - A) = 3$, find $n(A - B) + n(A \cap B) + n(B - A)$.

Options

A. 8

B. 5

C. 6

D. 9

Correct answer: 8

Explanation and steps

1. The governing definition or property gives: 8.

2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is 8.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 45 - A1.10(ii)

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Complete $A \cap B' = A - B$.

Options

A. $A \cap B' = \{1, 2, 3\}$

B. $A \cap B' = \{4, 5\}$

C. $A \cap B' = \{6, 7, 8\}$

D. $A \cap B' = \{1, 2, 3, 4, 5\}$

Correct answer: $A \cap B' = \{1, 2, 3\}$

Explanation and steps

1. The governing definition or property gives: $A \cap B' = \{1, 2, 3\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A \cap B' = \{1, 2, 3\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 46 - A1.10(iii)

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Complete $A' \cap B = B - A$.

Options

A. $A' \cap B = \{6, 7, 8\}$

B. $A' \cap B = \{1, 2, 3\}$

C. $A' \cap B = \{4, 5\}$

D. $A' \cap B = \{4, 5, 6, 7, 8\}$

Correct answer: $A' \cap B = \{6, 7, 8\}$

Explanation and steps

1. The governing definition or property gives: $A' \cap B = \{6, 7, 8\}$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A' \cap B = \{6, 7, 8\}$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.